

Title: math

Grade Level: 10

Subject: Mathematics

Difficulty: MEDIUM

Student Name: _____

Marks: 60 Pts

Time: 85 Mins

Roll No: _____ Section: _____

SECTION A: MULTIPLE CHOICE QUESTIONS

Instructions:

Choose the best matching option.

1. What is the value of x in the equation $2x + 5 = 11$?

[1 Marks]

2. Which of the following is equal to $\sin(30 \text{ degrees})$?

[1 Marks]

3. What is the formula for the area of a circle?

[1 Marks]

4. What is the value of $\cos(60 \text{ degrees})$?

[1 Marks]

SECTION B: SHORT QUESTIONS

Instructions:

Answer the questions in a brief paragraph.

1. Solve for x in the equation $x^2 + 4x + 4 = 0$.

[2 Marks]

2. Find the equation of the line that passes through points (2,3) and (4,5).

[2 Marks]

3. Solve the inequality $2x - 5 > 3$.

[2 Marks]

SECTION C: DIAGRAM/GRAPH-BASED QUESTIONS

Instructions:

Analyze and answer questions based on the given graph.

1. What is the x-intercept of the graph $y = x^2 + 2x - 3$?

[5 Marks]

2. Find the equation of the axis of symmetry for the graph $y = x^2 - 4x - 5$.

[5 Marks]

3. What is the maximum value of the function $f(x) = 2x^2 - 4x + 1$?

[5 Marks]

4. Determine the vertex form of the parabola with equation $x^2 + 6x + 8y + 1 = 0$.

[5 Marks]

5. Graph the equation $y = \text{abs}(x - 2)$ on a coordinate plane.

[5 Marks]

SECTION D: NUMERICAL PROBLEMS

Instructions:

Solve the numerical problems below.

1. A bakery sold 250 loaves of bread at \$2 each. What is the total cost?

[5 Marks]

2. A car travels 300 miles in 5 hours. What is its average speed?

[5 Marks]

3. A store is having a sale on TVs. Originally priced at \$800, a TV is on sale for 20% off. [5 Marks]
What is the sale price?

4. A group of friends want to split a bill evenly. If the total bill comes out to be \$120 and [5 Marks]
there are 8 friends, how much will each friend pay?

5. A water tank can hold 1000 gallons of water. If 300 gallons of water are already in [5 Marks]
the tank, what percentage of the tank is filled?

